LABORATORY RECORD

Course: Full Stack Web Development
Course Code: 23CM4121
Experiment No.: 4
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1. TITLE

JavaScript Basics – Arrays & Functions

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2. AIM

To understand and implement the basic concepts of **JavaScript arrays, functions, array methods, loops, and classes** by creating simple programs and observing their output.

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3. OBJECTIVE

The objectives of this experiment are:

- To understand the concept of arrays in JavaScript.
- To create and initialize arrays with values.
- To access and modify array elements using index numbers. To determine the length of an array.
- To use array methods such as `push()`, `pop()`, `shift()`, and `unshift()`.
- To iterate through array elements using a `for...of` loop.
- To understand the creation and use of functions in JavaScript.
- To pass arrays and values as function arguments.
- To add and remove elements from an array using functions.
- To understand classes, constructors, methods, and private fields in JavaScript.

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4. THEORY

JavaScript is a high-level programming language commonly used to add dynamic behavior and interactivity to web applications. It supports various programming concepts such as variables, arrays, functions, objects, classes, loops, and methods.

Arrays

An **array** is a collection of multiple values stored in a single variable. Array elements are accessed using an index, and the index starts from **0**.

For example: `const colors = ["Red",`

`"Green", "Blue"];`

Here, `Red` is at index `0`, `Green` is at index `1`, and `Blue` is at index `2`.

JavaScript provides several built-in array methods:

- `push()` – adds an element to the end of an array.
- `pop()` – removes the last element.
- `shift()` – removes the first element.
- `unshift()` – adds an element to the beginning.
- `length` – returns the number of elements in an array.

Arrays can also be processed using loops such as the `for...of` loop.

Functions

A **function** is a reusable block of code designed to perform a particular task. Functions can accept parameters and can be called multiple times with different values.

In this experiment, functions are used to:

- Display shopping cart items.
- Add a new item to the cart.
- Remove the last item from the cart.

Classes and Private Fields

JavaScript also supports **classes**, which provide a way to create objects with properties and methods. A constructor is used to initialize an object.

The `#` prefix is used to declare a **private class field**. A private field cannot be accessed directly from outside the class. In the BankAccount program, `#balance` is private and is accessed through the `getBalance()` method.

The basic concepts demonstrated in this experiment are:

Arrays → Array Methods → Loops → Functions → Function Calls → Classes → Private Fields

5. ALGORITHM / PROCEDURE

1. Create a JavaScript project or folder.
2. Create the required JavaScript files.
3. Create an array containing initial values.
4. Display the original array using `console.log()`.
5. Access individual array elements using index numbers.
6. Find the total number of elements using the `length` property.
7. Modify an element using its index.
8. Add an element using `push()`.
9. Remove the last element using `pop()`.
10. Remove the first element using `shift()`.
11. Add an element to the beginning using `unshift()`.
12. Iterate through the array using a `for...of` loop.
13. Create reusable functions for shopping cart operations.
14. Pass the shopping cart array as a function argument.
15. Add and remove items using separate functions.
16. Create a `BankAccount` class with a private balance field.
17. Create a constructor to initialize the account owner.
18. Create methods to deposit money and retrieve the balance.
19. Execute the programs using a JavaScript runtime/browser console.
20. Observe and record the output of all three programs.

6. PROGRAM

Program 1: array.js

```
// 1. Creating an array with initial values const
colors = ["Red", "Green", "Blue"];
console.log("Original List:", colors);

// 2. Accessing items using index numbers (Starts at 0)
console.log("First color (index 0):", colors[0]); // Output: Red
console.log("Second color (index 1):", colors[1]); // Output: Green

// 3. Finding the total number of items
console.log("Total colors in list:", colors.length); // Output: 3
```

```
// 4. Updating an item at a specific index colors[1] =
"Yellow"; // Changes "Green" to "Yellow"
console.log("After updating index 1:", colors);

// 5. Adding a new item to the END of the array colors.push("Purple");
console.log("After adding to the end (.push):", colors);

// 6. Removing the last item from the array
const removedColor = colors.pop(); // Removes "Purple" and saves it
console.log("Removed color:", removedColor);
console.log("Array after removing last item (.pop):", colors);

const remove = colors.shift(); // Removes 1st element from array
console.log("Remove:",remove)

colors.unshift("Pink") //Adds element at 1st
console.log("adding at start",colors)

// 7. Looping through the array to see every item
console.log("\n--- Printing all colors one by one ---"); for
(const color of colors) {
   console.log(`Color: ${color}`);
}
```

Program 2: oop.js

```
class BankAccount {
 // 1. Declare the private field using the '#' prefix
 #balance = 0;

 constructor(owner) {
  this.owner = owner; // Public property
 }

 // 2. Control access via a public method (Getter)
getBalance() {     return this.#balance;
 }

 // 3. Validate modifications via a public method (Setter-like operation)
deposit(amount) {     if (amount > 0) {       this.#balance += amount;
     console.log(`Deposited ${amount}.`);
   } else {
     console.log("Invalid deposit amount.");
   }
```

```
  }
}

// --- Usage ---
const myAccount = new BankAccount("Alex");

myAccount.deposit(500);       // Output: Deposited 500.
console.log(myAccount.getBalance()); // Output: 500
```

Program 3: script.js

```
const shoppingCart = ["Laptop", "Headphones", "Mouse Pad"];
// Function to print all items currently in the cart function
viewCart(cartArray) {
   console.log(`\n Your Cart (${cartArray.length} items)`);
if (cartArray.length === 0) {        console.log("Your cart is
empty!");
       return; // Stops the function early if cart is empty
   }
   for (let i = 0; i < cartArray.length; i++) {
      console.log(`${i + 1}. ${cartArray[i]}`);
   }
}
// Function to add a new item to the cart function
addItem(cartArray, item)
{    cartArray.push(item);
   console.log(`Added to cart: ${item}`);
}
// Function to remove the very last item added
function removeLastItem(cartArray) {     if
(cartArray.length > 0) {        const removed =
cartArray.pop();
       console.log(`Removed from cart: ${removed}`);
   } else {
      console.log("Nothing to remove. Cart is already empty!");
   }
}
// 3. RUNNING THE PROGRAM (Calling the functions)
viewCart(shoppingCart);       // Shows the initial 3 items
addItem(shoppingCart, "Keyboard"); // Adds a 4th item
viewCart(shoppingCart);       // Shows updated list
removeLastItem(shoppingCart); // Removes "Keyboard"
viewCart(shoppingCart);       // Shows final list
```

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7. CODE EXPLANATION

Code / Concept	Explanation
const colors = [...]	Creates an array containing initial color values.
colors[0]	Accesses the first element of the array.
colors.length	Returns the total number of elements in the array.
colors[1] = "Yellow"	Updates the element at index 1.
push()	Adds a new element to the end of an array.
pop()	Removes and returns the last element of an array.
shift()	Removes and returns the first element of an array.
unshift()	Adds a new element to the beginning of an array.
for...of	Iterates through each element of an array.
console.log()	Displays output in the console.
class BankAccount	Defines a class named `BankAccount`.
#balance	Declares a private class field.
constructor()	Initializes an object when it is created.
this.owner	Stores the account owner's name as a public property.
getBalance()	Returns the value of the private balance field.
deposit()	Adds a valid amount to the account balance.
if	Performs conditional checking.
new BankAccount()	Creates a new object from the `BankAccount` class.
function viewCart()	Defines a function to display shopping cart items.
function addItem()	Defines a function to add an item to the cart.
function removeLastItem()	Defines a function to remove the last cart item.
return	Exits the function when the cart is empty.
Function parameter	Allows data such as the cart array or item to be passed into a function.
cartArray.push(item)	Adds an item to the shopping cart.
cartArray.pop()	Removes the last item from the shopping cart.

8. TEST CASES AND EXECUTION DATA

The three JavaScript programs were executed and their console outputs were verified.

Program 1: Array Operations

Test Case	TC-01

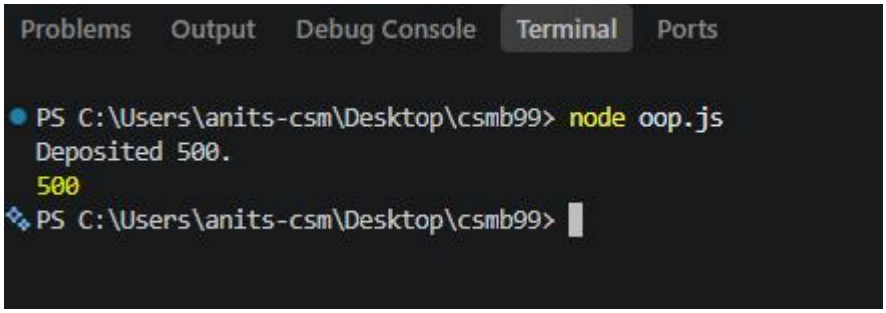
Test / Input	Execute `array.js` using Node.js
Expected Result	The program should create an array of colors and demonstrate indexing, length calculation, updating an element, adding elements using `push()`, removing elements using `pop()` and `shift()`, and displaying all colors using iteration.
Actual Result	The program successfully displayed the original list, indexed colors, total length, updated list, added Purple, removed Purple, removed Red, added Pink, and printed all colors one by one.
Status	**Pass**

TC-01 Output:

Program 2: BankAccount Class

Test Case	TC-02
Test / Input	Execute `oop.js` using Node.js
Expected Result	The BankAccount class should successfully create a bank account, deposit `500`, and display the updated account balance.
Actual Result	The program successfully displayed `Deposited 500.` and the account balance as `500`.
Status	**Pass**

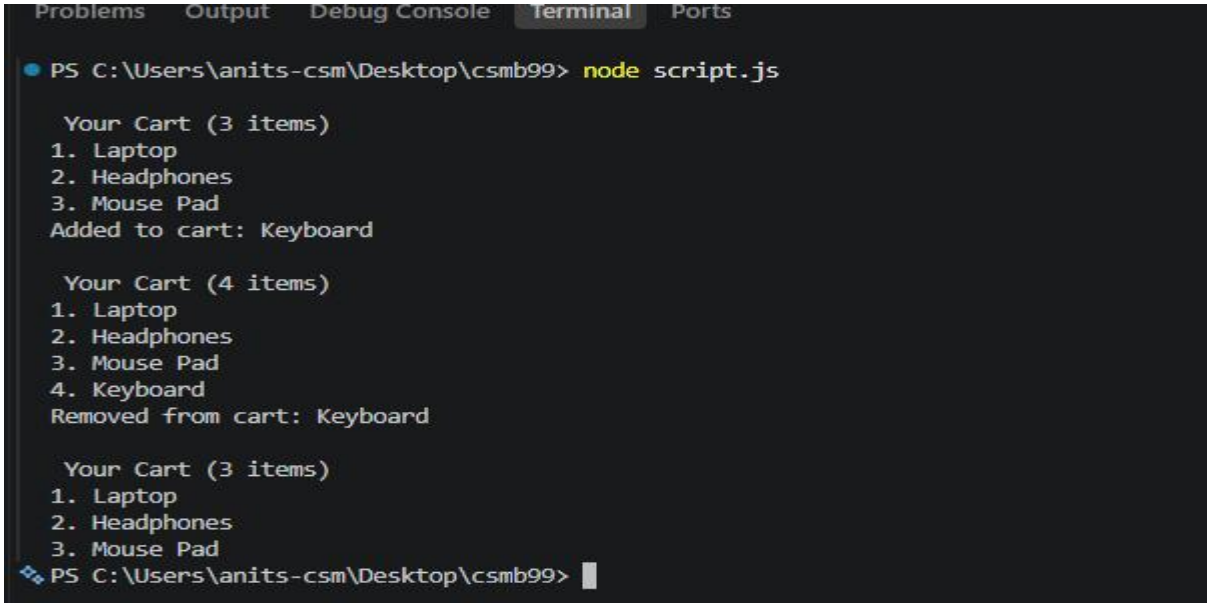
TC-02 Output:



Program 3: Shopping Cart Functions

Test Case	TC-03
Test / Input	Execute `script.js` using Node.js
Expected Result	The shopping cart should initially display 3 items, add `Keyboard` to the cart, display the updated cart with 4 items, remove `Keyboard`, and finally display the cart with the original 3 items.
Actual Result	The program successfully displayed the initial cart, added Keyboard, displayed the updated 4-item cart, removed Keyboard, and displayed the final 3-item cart.
Status	**Pass**

TC-03 Output:



9. OUTPUT

The three JavaScript programs were executed successfully. The corresponding console outputs are shown below.

Output 1: Array Operations – array.js

```
PS C:\Users\anits-csm\Desktop\csmb99> node array.js
Original List: [ 'Red', 'Green', 'Blue' ]
First color (index 0): Red
Second color (index 1): Green
Total colors in list: 3
After updating index 1: [ 'Red', 'Yellow', 'Blue' ]
After adding to the end (.push): [ 'Red', 'Yellow', 'Blue', 'Purple' ]
Removed color: Purple
Array after removing last item (.pop): [ 'Red', 'Yellow', 'Blue' ]
Remove: Red
adding at start [ 'Pink', 'Yellow', 'Blue' ]

--- Printing all colors one by one ---
Color: Pink
Color: Yellow
Color: Blue
PS C:\Users\anits-csm\Desktop\csmb99>
```

Figure 1: Output of Array Operations using JavaScript

The output demonstrates array creation, indexing, length calculation, updating, adding, removing, and looping operations.

Output 2: Bank Account Class – oop.js

```
Problems  Output  Debug Console  Terminal  Ports

PS C:\Users\anits-csm\Desktop\csmb99> node oop.js
Deposited 500.
500
PS C:\Users\anits-csm\Desktop\csmb99>
```

Figure 2: Output of BankAccount Class using JavaScript

The output shows successful deposit of `500` into the bank account and retrieval of the account balance using a public method.

Output 3: Shopping Cart Functions – script.js

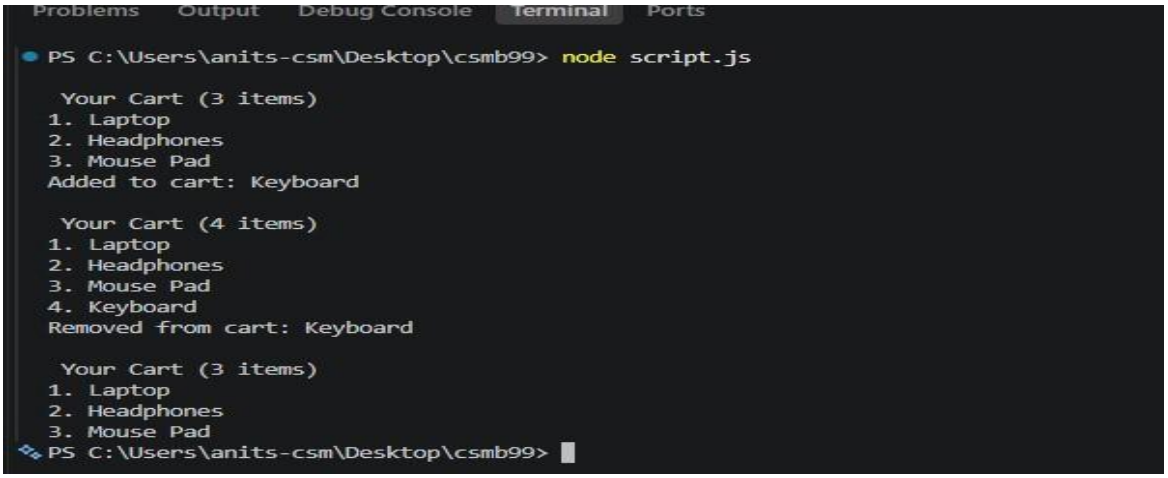


Figure 3: Output of Shopping Cart Operations using JavaScript Functions

The output demonstrates displaying the initial cart, adding an item, displaying the updated cart, removing the last item, and displaying the final cart.

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10. RESULT

Thus, the basic concepts of **JavaScript arrays and functions** were successfully implemented and executed. Array operations such as indexing, updating, push(), pop(), shift(), unshift(), and iteration were demonstrated. Functions were used to perform shopping cart operations, and a JavaScript class with a private field was implemented using the # syntax. All three programs were executed successfully and the expected outputs were obtained.